

PAK-ANGELS MID PROGRAM HACKATHON

FreshSense

Automated Food Freshness Diagnostic Platform Powered by Deep
Learning & Computer Vision

Model: **MobileNetV2**

Validation: **82% F1 Score**

Granularity: **5 Freshness Stages**

Execution: **Zero GPU / Local CPU**

What critical issue are we solving?

Global produce wastage and foodborne risks stemming from subjective, unreliable manual inspection.

01 • GLOBAL IMPACT

Massive Food Waste

Over 1/3 of all perishable food is lost annually. In retail and domestic kitchens, lack of early detection leads to premature binning of edible produce.

**\\$1.3
Trillion**

Annual global food
loss

02 • SENSORY FLAWS

Subjective Estimation

Manual human evaluation relying on smell, touch, and sight is inconsistent, varies by handler, and frequently misses internal biological breakdown.

**40%
Error**

In subjective spoilage
checks

03 • BINARY SHORTCOMING

Binary Fallacy

Existing tools categorize produce as either "Fresh" or "Rotten", completely failing to flag early degradation stages where salvage or immediate cooking is optimal.

**Missing
Stages**

No transitional stage
alerts

How does FreshSense solve this?

Instant, non-invasive computer vision that grades produce quality across five biological degradation stages.



Instant Photo Diagnostics

Drag-and-drop or snapshot any fruit/vegetable.
Evaluates features in under 850 milliseconds locally.



Actionable Culinary Recommendations

Gives clear consumption guidance based on stage (e.g. raw consumption vs. immediate stewing vs. discard).



Audit Log & Diagnostic History

Chronological scan log with timestamps and confidence records to track produce degradation over time.

5-STAGE QUALITY CLASSIFICATION SPECTRUM

1. Fresh

Harvest Grade • Raw Consumption

2. Slightly Aged

Minor Dehydration • Eat in 48h

3. Stale

Cook / Puree • Avoid Raw

4. Spoiled

Early Microbial • Do Not Consume

5. Rotten

Severe Decomposition • Bio-Waste

Engineered with a robust modern stack.

End-to-end integration leveraging cutting-edge agentic workflows, deep learning, and resilient web services.



PLATFORM & WORKFLOW

Google Antigravity

Autonomous agentic pairing environment facilitating architecture exploration, code refactoring, and rapid verification loops.

Agentic AI

Pair Programming



MACHINE LEARNING

MobileNetV2

Depthwise separable convolutions trained with Agglomerative Clustering labels. High F1 score (0.82) packed in just 11.5 MB.

Keras 3

NumPy Engine

Pillow



BACKEND INFRASTRUCTURE

Python 3.14 & Flask

Lightweight REST API orchestrating image preprocessing, silent thread-safe inference, and JSON persistence.

Flask 3.1

Silent Mode

JSON DB



FRONTEND & STYLING

Modern Design System

Figma-inspired editorial UI with Outfit & Work Sans typography, animated probability bars, and zero bulky framework overhead.

HTML5 / CSS3

Vanilla ES6

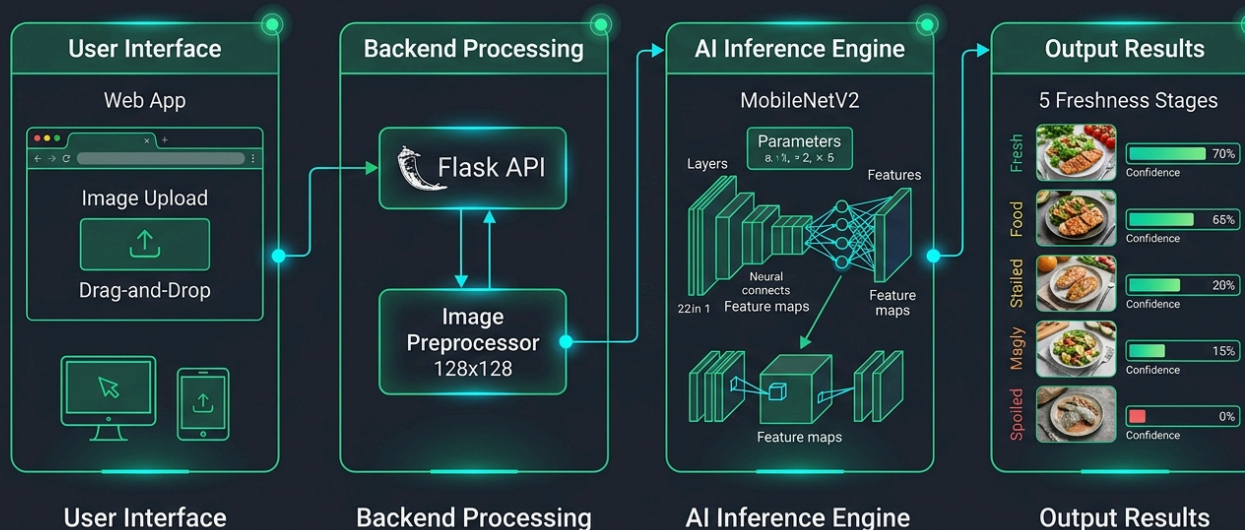
SVG Icons

High-Level System Communication Diagram

End-to-end data pipeline from user ingestion to neural network classification and persistent storage.

FreshSense': High-level Software System Architecture Diagram

AI food freshness detection system



Meet the Team Behind **FreshSense**

Cross-functional collaboration across machine learning, system design, and product engineering.

MF

Muhammad Faseeh (GL)

GROUP LEADER

Project Strategy, AI Pipeline
Integration & Backend Orchestration

UI

Umamah Ibreeq Zafar

TEAM MEMBER

Computer Vision Research, Dataset
Preparation & Model Evaluation

AN

Anumta Nadeem

TEAM MEMBER

UI/UX Design Systems, Client
Interactions & Responsive Interface

ZA

Zafar Aman Khattak

TEAM MEMBER

System Architecture, Diagnostic
Logging & Quality Assurance

Thank You! Questions & Live Demonstration

FreshSense • Reducing food waste with intelligent computer vision.

• SUBMISSION COMPLETE